

Clinical Impact of Pre-Transplant Microbial Diversity on Transplant Outcomes

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Abstract

Introduction

In allogeneic stem cell transplantation (allo-SCT), recent reports have shown that post-transplant microbial diversity of the gastrointestinal tract is closely associated with clinical outcomes. Furthermore, pre-transplant microbial status has also been associated with the development of acute graft-versus-

host disease (GVHD). These results indicate that pre-transplant microbial ecosystem may influence the immune system after allo-SCT, induce alloimmune response such as GVHD, and may finally affect the transplant outcome. Thus, we evaluate the association between pre-transplant microbial diversity and transplant outcomes.

Patients and Methods

Between April 2013 and March 2015, fecal samples were obtained from 107 patients in Komagome hospital. Fecal samples were collected within 2 weeks before conditioning. Microbial analysis was performed using 16S rRNA gene (hypervariable V1-2 region) sequencing. Operational taxonomic units (OTU)-based microbial diversity was estimated by calculating the Shannon index. Patients were classified into 3 groups based on the diversity index: low (<2), intermediate (2-3), and high (>3) diversity group, and we evaluated transplant outcomes such as survival, non-relapse mortality, and the cumulative incidence of GVHD, in these 3 groups.

Results

Among the 107 patients, there were 18 (16.8%) patients in the low diversity group, 48 (44.9%) in the intermediate diversity group, and 41 (38.3%) in the high diversity group. The median age of all patients was 49 (range: 16-72) years. The median follow-up period for survivors was 597 days (range, 23-1,046 days). We found no significant differences among 3 groups in terms of age, gender, underlying disease, disease status, performance status, hematopoietic cell transplantation comorbidity index, GVHD prophylaxis, conditioning regimen, stem cell sources, and human leukocyte antigen disparity. However, the patients in the low diversity group received intravenous antibiotics just before the conditioning more frequently compared to the patients in the intermediate /high diversity groups.

The patients in the low diversity group showed poor survival ($p=0.37$, Figure 1a), higher non-relapse mortality ($p=0.25$, Figure 1b) and higher incidence of grade II-VI gastrointestinal GVHD ($p=0.77$, Figure 1c) compared with the patients in the intermediate /high diversity group although these results were not statistically significant. The cumulative incidence of relapse was similar between 3 groups. When compared between patients with and without acute GVHD, the composition of microbiota did not differ significantly between these patients.

Discussion and conclusion

Our results indicate that pre-transplant microbial diversity did not affect transplant outcomes. Furthermore, in contrast to previous report, the microbial composition was also irrelevant to the development of GVHD. Bacterial translocation after allo-SCT is a key trigger for the development of acute GVHD. Thus, microbial status in this period would be more important than that before allo-SCT. However,

the patients in the low diversity group tended to show poor survival, higher incidence of non-relapse

mortality and GVHD. Thus, further evaluation with larger sample size might be necessary to confirm the association between pre-transplant microbial ecosystem and transplant outcomes.

Figure 1a: Overall Survival

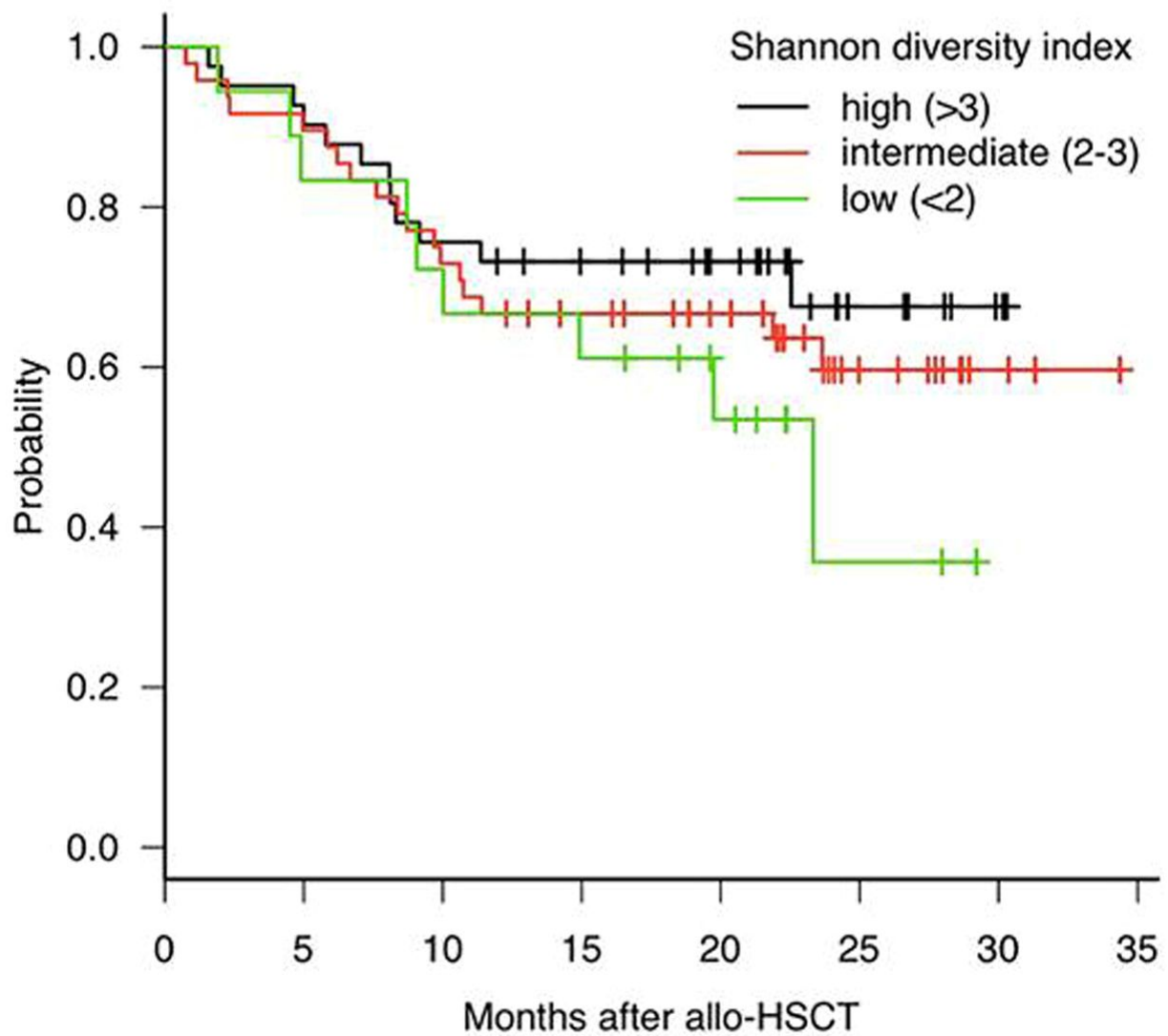


Figure 1b:
Cumulative incidence of non-relapse mortality



Disclosures

No relevant conflicts of interest to declare.

Author notes

*Asterisk with author names denotes non-ASH members.

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